Edit

GSP 426

Mapping & Data Visualization

Spring 2026

Welcome to Mapping & Data Viz!

Maps are so often treated as authoritative, yet we seldom realize how much of the message is influenced by choices made by the cartographer. In this short-format course, we'll explore various components of geographic data display. You may be familiar with many of them, but we'll take a deeper dive into meaning and intent as we work through data selection, design choices, and cartographic techniques, evaluating what is revealed, and what remains hidden.

This practicum explores how geographic data can be transformed into compelling, ethical, and analytically sound visual stories, and how those stories shape understanding, decision-making, and power. We will critically examine cartographic design, spatial narratives, and the social implications of maps while gaining hands-on experience building spatial stories with modern geovisualization tools. Emphasis is placed on spatial thinking, storytelling structure, audience awareness, and ethical responsibility in geographic communication.

About the Course

Course Details:

- **Name**: GSP 426 Mapping & Data Visualization (Cartography Practicum)
- **Term**: Spring 2026
- **Sections**: 23662

Course Format: This 1-unit course will meet in person for the first 8 weeks, with meeting days and times noted in your student center. The rest of the course is considered online asynchronous, which for our purposes will consist of the final project and related critique. All assignments will be submitted in Canvas. **Keep track of due dates in the Canvas Calendar, or by using the To Do list on your Dashboard or course homepage.**

Expected Work Load: The expected workload at Cal Poly Humboldt is calculated at 3 hours per week for each unit in a standard 16-week course. For this 1-unit practicum course, this amounts to 3 additional hours per week, outside of class meeting times. Since we are front-loading the course into the first half of the term, two hours per week are face-to-face classroom work, and then up to 6 hours completing assignments on your own time. Some weeks, the load may be lighter than others, but you should always plan on setting aside this time. The expectation is that all components will be completed by week 9, with no in-person meetings after week 8.

Credits: 1 unit; repeatable for credit

Catalog Description: Explore spatial thinking through hands-on mapping and visualization techniques. Integrate geospatial technologies with applied projects to analyze, interpret, and communicate patterns across space, place, and data.

Prerequisites: GSP 270 or 272 (GIS) and GSP 316 (Cartography) are **strongly recommended** before taking this course. You should be comfortable with finding, retrieving, and manipulating data to get it into your map.

Course Objectives

After completing this course, students will be able to:

- Critically evaluate geographic visualizations
- Design maps and data stories with intentionality
- Integrate spatial and non-spatial evidence
- Communicate complex geographic ideas clearly and ethically

This course is a requirement for several degree programs, and it contributes to the achievement of Program Learning Outcomes (PLOs) and Institutional Learning Outcomes (ILOs).

Cal Poly Humboldt Institutional Learning Outcomes (https://www.humboldt.edu/academic-programs/cal-poly-humboldt-undergraduate-institutional-learning-outcomes)

Required Materials

Course Text

All readings will be provided for this course. However, you may wish to acquire some of these foundational texts for your own collection.

- **Required (but provided):**
  - ***How to Lie with Maps***, by Mark Monmonier

  - ***How Charts Lie***, by Alberto Cairo
- **Recommended (select chapters provided)**:
  - ***Thematic Mapping,*** by Kenneth Field
  - ***Mapping Time***, by Menno-Jan Kraak
  - ***The Visual Display of Quantitative Information***, by Edward Tufte

Additional readings may be provided or assigned, in the form of materials available freely online or supplied PDFs.

Course Format

This course is organized into a series of modules, which group together a series of related readings, activities, and assignments. Each module may be accessed from the course home page, or will usually be linked in the weekly announcements.

The course is organized into **eight in-person weeks**, followed by a final project submission. Each week integrates:

- Conceptual lecture and discussion
- Guided, low-stakes technology practice
- Independent application using student-selected data
- Critical writing and reflection

Assignments are due typically one week from the session in which they are assigned. Examinations and assignments are listed in the course outline below, and are **subject to change**. You are responsible for knowing when your work is due – any changes to due dates will be posted on the Canvas calendar and in Announcements. Please see your Canvas calendar and/or the Syllabus link for the most current version.

Technical Requirements

- **Primary spatial platform:** ArcGIS Online
- **Statistical graphics:** Excel (for data hygiene and supplemental graphics)
- **Design tools (optional):** Adobe Illustrator / InDesign
- **Other platforms (optional):** Various online mapping/visualization applications

If you wish to use ArcGIS Pro or Illustrator on your own computer, it is your responsibility to make sure it meets requirements. For more information, see the **Software for Home Use (https://hsu-forms.humboldt.edu/software/)** page.

Accessing the Software

This course is primarily taught using Esri's **ArcGIS Online (https://humboldtstate.maps.arcgis.com/)**, which is readily accessible from any internet-connected computer. Detailed instructions for login are **here (https://its.humboldt.edu/arcgis-instructions#ago/)**.

You may also use ArcGIS Pro for assignments where appropriate. Student versions may be downloaded from the **Software for Home Use (https://hsu-forms.humboldt.edu/software/)** page and selecting **ArcGIS Pro**. License information is contained in the **Installation instructions**.

Please note, **ArcGIS Pro is only compatible with the Windows operating system**. If you have a Mac or other non-Windows device, you may access the software from the **Virtual Lab (https://canvas.humboldt.edu/courses/82139/pages/accessing-the-software)**.

Computers on Campus

As an enrolled student, you are welcome to use any of the geospatial computer labs on campus during open lab hours. These labs include Forestry 203 and 204A, and Natural Resources 203. There are also computers on the first floor of the University Library with geospatial software installed. Students may inquire at the Library Help Desk for details on the current status of computers and software.

Courses are also taught in these computer labs, so **please be respectful.** Check the schedule posted on the door before entering. **Do not** interrupt an ongoing class. If you arrive at the start of a class, always ask the instructor permission to use the computers if there is a class in session.

Information on lab locations and availability can be found on the **ITS Computer Labs webpage (https://its.humboldt.edu/classrooms/labs)**.

Course Schedule

The recommended method for navigating this course is via the home page, and then working through each module from top to bottom. Specific due dates are listed with each activity, and can also be found on the Canvas Calendar, your To Do list, and the bottom of this page.

Module 1: Maps as Stories, Maps as Power

Maps are not neutral artifacts, but rather authoritative stories.

- Maps as arguments: what is emphasized, omitted, or normalized
- Audience and intent in geographic communication

Module 2: Seeing Patterns (and Missing Them)

Looking into the ethical implications of aggregation and representation.

- Scale, resolution, and extent
- Projection as interpretation
- Modifiable Areal Unit Problem (MAUP)

Module 3: Lies, Damned Lies, and Statistical Maps

With the right (or wrong) technique, a map can prove almost anything.

- Classification as a modeling choice
- Statistical vs. visual patterns: “looks clustered” vs. “is statistically clustered”
- Correlation and causation, and spurious pattern-finding

Module 4: Color, Perception, and Accessibility

Beyond color as a visual variable, and into color as persuasion or barrier.

- Color as a visual variable (hue, value, saturation)
- Visual hierarchy and emphasis
- Accessibility: color acuity, contrast, and legibility

Module 5: Temporal Mapping, Change, and Narrative

Time shows us what changed but also sometimes why and how much.

- Time as a narrative device
- Temporal framing and implied causality
- Techniques for mapping change

Module 6: When Maps Aren’t Enough

Sometimes the map doesn’t tell the whole story.

- Maps vs charts: what’s the best tool for the job?
- Supporting vs competing narratives
- Overload, redundancy, and omission

Module 7: Putting the Story Together

A data story is a structured narrative, not a collection of visuals.

- Integration and consistency of visuals (harmony and unity)
- Visual hierarchy and reading order
- Layout as argument: moving the reader through the story

Module 8: Using Your Power Wisely

Centering power, information, and responsibility in spatial storytelling.

- Creating a compelling spatial narrative
- Providing constructive peer critique
- Incorporating feedback and evaluating results

Your Instructor

Meet the Instructor

Amy E. Rock, Ph.D., GISP
Department of Geography, Environment & Spatial Analysis
Founders Hall 131
amy.rock@humboldt.edu (mailto:amy.rock@humboldt.edu)

Preferred name: Please call me Doc Rock!

Pronouns: she/her

How to reach me: The best way to connect with me is via Slack (DM for personal matters), as I get tons of emails, especially at the start and end of a semester, which might mean that I don't see your message until it's too late.

A little more about me

Dr. Amy Rock is a longtime resident of Ohio, and a relatively recent transplant to Northern California, where she teaches geography, GIS and cartography in the Department of Geography, Environment, and Spatial Analysis at Cal Poly Humboldt. She is also the Executive Director of the University Consortium for Geographic Information Science (http://www.ucgis.org), and a member of the editorial board of Cartographic Perspectives (https://cartographicperspectives.org/index.php/journal).

Doc Rock encourages her students to become engaged in their communities and think critically about politics, economic and community development and the role of

NOTE: I will read and respond to e-mail and Slack posts at least once per day, Monday through Friday.

nonprofit organizations in our society. She works with nonprofits and community organizations to bring geospatial thinking to contemporary issues. An avid map geek, she shamelessly judges a book (or video game) by its map.

Professional Affiliations

Doc Rock belongs to several cartography-related organizations, many of which have student membership options. These organizations can help you get connected in the professional community, as well as see what cool stuff people are doing. Several have annual conferences with map galleries, and map competitions with cash prizes. Check them out!

- **North American Cartographic Information Society (NACIS)** **(https://nacis.org/)**
- **Cartography and Geographic Information Society (CaGIS)** **(https://cartogis.org/)**
- **Cartography Specialty Group** **(https://aagcartography.wordpress.com/)** of the American Association of Geographers (AAG)
- Esri Cartography Special Interest Group (CartoSIG)
- **International Cartographic Association (ICA)** **(https://icaci.org/)**
- **California Map Society** **(https://californiamapsociety.org/)**
- **Women in GIS (WiGIS)** **(https://womeningis.wildapricot.org/)**

Course Communication & Professional Conduct

This is an upper-division course. Students are expected to communicate clearly, professionally, and respectfully in all course-related interactions, including Canvas, Slack, email, and peer critique.

Primary Communication Channels

- **Canvas Announcements** are the official channel for course updates. You are responsible for reading them.
- **Slack** is the primary space for questions about assignments, tools, or course concepts. If the question is not personal or confidential, it belongs in Slack.
- **Email** should be used for personal, confidential, or grade-related matters only. Use your Cal Poly Humboldt email account.

Professional Communication Expectations

- Identify yourself clearly (name, course, section) when emailing.
- Use a meaningful subject line (e.g., “GSP 426 – Final Project Question”).

- Write in complete sentences using a respectful, professional tone.
- Be concise and specific — provide enough context for your question.
- Check Canvas materials and Slack history before asking a question.

Communication that is unclear, disrespectful, or dismissive of course expectations may receive delayed responses or be redirected to the appropriate channel.

Online Conduct & Critique Culture

Critique and discussion are central to this course. Address ideas, designs, and arguments — not people. Disagreement is expected; disrespect is not.

- Be constructive, specific, and grounded in course concepts.
- Avoid sarcasm, inflammatory language, or personal attacks.
- Assume good faith, but take responsibility for your words.
- Maintain confidentiality and discretion when discussing sensitive data or topics.

Standards for communication in this course align with professional expectations in cartography, GIS, and data visualization practice.

Evaluation and Grading

Grade Mode

This course is optional grade mode — it may be taken for a letter grade or as credit/no-credit (CR/NC). Students must select the grade mode in their Student Center by the deadline indicated on the **academic calendar (https://www.humboldt.edu/events/academic-deadlines)**.

To count towards fulfilling major requirements, this course must be taken for a letter grade. Before choosing CR/NC mode, review the Registrar's guidance: **CR/NC and grade mode information (https://registrar.humboldt.edu/grades)**.

How Your Grade Is Determined

Your grade is based on the following course components:

- Reading syntheses (20%)
- Weekly mapping activities (35%)

Grading Scale

Grade	Range
A	93% to 100%
A-	90% to < 93%
B+	87% to < 90%
B	83% to < 87%
B-	80% to < 83%
C+	77% to < 80%
C	73% to < 77%
C-	70% to < 73%

- Final project proposal (10%)
- Final project (35%)

All grades will be posted in Canvas. Feedback is provided via rubrics and comments — please read it carefully, as it is designed to help you improve your work on the next item. Assignments are typically graded within one week of submission (depending on instructor workload). You can track your progress in the Canvas **Grades** area throughout the term.

Participation and Professionalism

Participation does not mean speaking constantly. It means thoughtful contribution to critiques and discussions, asking good questions, and supporting your peers in constructive ways. There are multiple ways to demonstrate engagement, including written feedback and attentive critique participation.

Due Dates and Updates

Due dates are listed in Canvas. Any changes will be announced via Canvas Announcements and reflected in the Canvas Calendar and assignment pages. Canvas due dates should be considered the most current.

Late Policy

Each activity builds on the one before it, so it's important to stay on track. Late submissions will automatically display as a **0 in Canvas** until submitted and graded.

- **Late work is penalized 20% per day.**
- In-class interactive activities may not be submitted late or made up without an excused absence.

If something significant happens, communicate as early as you can (no need for full details; just the basics). In some cases, flexibility may be possible. Plan travel and personal events around your schoolwork, not vice versa.

Technical difficulties (computer issues, internet outages, power outages) do not automatically

Grade	Range
D+	67% to < 70%
D	63% to < 67%
D-	60% to < 63%
F	0% to < 60%

Letter grades are assigned using the following percentage ranges.

constitute an exception. If you experience issues while working from home, come to campus or use an alternative location such as a local library. In the case of a widespread outage, deadlines will be adjusted.

Exceptions are granted only in cases of legitimate excused absences (e.g., medical documentation, court summons, deployment notice). An excused absence means you have either (a) notified the instructor in advance and received confirmation or (b) provided written documentation. The more advance notice you provide, the more flexible we can be.

Incompletes

An authorized Incomplete indicates that a portion of required coursework has not been completed and evaluated in the prescribed time period due to unforeseen, but fully justified, reasons and that there is still a possibility of earning credit (Office of the Registrar). Incompletes are not a “fail-safe” for students who have neglected to submit sufficient work to pass the course. An incomplete will not be considered for students who have not completed a significant portion of the work prior to the circumstances justifying the incomplete. It is the student’s responsibility to bring pertinent information to the instructor’s attention and discuss the terms of an incomplete contract.

Policy on Generative AI

Generative artificial intelligence (AI) programs, such as ChatGPT, may be used in this course to brainstorm or learn more about a topic, summarize readings, revise your work, draft an outline or organize thoughts, check grammar, or improve writing style with appropriate citation complying with Humboldt's Student Academic Honesty Procedure. You are responsible for fact checking statements composed by generative AI models and respecting intellectual property (sometimes it's just plain wrong!). Generative AI may not be used for composing discussion posts or completing assignments, writing papers/reports or portions of papers/reports for class assignments. If you are uncertain about the appropriate use of generative AI in this course, you are encouraged to contact your instructor for clarification. Inappropriate use of generative AI programs is not permitted and will be treated as plagiarism as defined in Humboldt's Student Academic Honesty Procedure and handled in accordance with the Procedure.

Meeting Minimum Requirements

Life gets busy, and there will be times where you have a lot going on. It may be tempting to submit assignments that do not meet minimum requirements (including word count). Work that barely addresses requirements risks significant point loss, including a 0 for a sloppy or wholly inadequate

submission. In general, it's best to fully articulate your ideas and maintain a safety margin for both word count and timely submission.

Meeting Minimum Word Count Requirements

- Submitting only the minimum word count typically earns at most a **C** (unless exceptionally concise and strong).
- To earn an **A**, exemplary work should exceed the minimum requirements in a meaningful way (avoid "word salad").
- The **main body text** determines word count; title page, headers/footers, tables of contents, and bibliography do not.
- Work well below the minimum word count may receive a **0**.

If you're stuck on how to elaborate without padding, imagine explaining your post or report to a friend. What questions would they have? For lab-style work, could someone reproduce what you did (assuming software knowledge), or do they need more detail about choices and results? While submitted work must be your own, it's okay to talk ideas through with a peer to ensure your writing says what you think it says.

Course Policies

Attendance Policy and Student Expectations

Your attentive presence is expected at all class sessions (for in-person sections), as is participation in class discussions and activities. The collaborative nature of critiques and other aspects of the course are hard to capture without being here. If you are unable to attend class, it is your responsibility to get the information covered in class. Regular patterns of absence will also adversely impact your performance in other areas, so come to class and learn stuff! (In an online course, this equates to watching the lectures regularly.)

Discussions in class and online are meant to enrich the knowledge of us all, and you are encouraged to ask questions, and share your perspectives and experiences. Disruptive behavior will not be tolerated. You are expected to be respectful to your instructor and your fellow classmates, in both active discussion and general demeanor. Phones, mp3 players and other distractions should be turned off and out of sight during class, except as needed for quizzes. Laptops and tablets may be used for notetaking, so long as their use is not distracting. If you are arriving late or need to leave early, please do us the courtesy of informing your instructor in advance, sit near the door and be respectful of others as you arrive/depart.

Please also avoid wearing/smelling of strong scents, as these can induce migraines in those of us who are susceptible. Perfumes, body spray, tobacco and alternatives, and even fabric softener can be overwhelming to those around you.

Academic Integrity

As noted elsewhere, cartography is a collaborative endeavor, and it is expected that you and your peers will solicit and provide advice and critique. You are still responsible for completing and submitting your own work, and work which draws heavily from the work of another without adequate citation may be considered plagiarism. This course follows the Cal Poly Humboldt guidelines for academic integrity. Students are responsible for knowing the policy regarding academic honesty: **Academic Honesty Policy (http://humboldt.edu/studentrights/academic-honesty)** .

Academic dishonesty is a severe breach of trust upon which the success of this course depends. As such, this course maintains a **zero tolerance** policy for any violations of academic integrity. Students who violate Humboldt State University's academic honesty policy will receive a failing grade in the course and may be held in violation of the University's code of conduct.

Students with Disabilities

We are dedicated to helping accommodate students with disabilities, and I have designed this course with ADA compliance in mind. If there are additional disability-related accommodations needed, students should contact the Student Disability Resource Center in House 71, 826-4678 (voice) or 826-5392 (TDD). Some accommodations may take up to several weeks to arrange. Please bring us your accommodation letter as soon as possible, so that we know how to meet your needs.

Campus Disability Resource Center Website (http://www.humboldt.edu/disability)

Related Policies and Procedures:

Student Records and Confidentiality (http://disability.humboldt.edu/studentservices-confidentiality)

Service Dog Policy (https://www.humboldt.edu/policies/animals-campus-policy-student-campus-residents-and-visitors)

A complete list of services available to support your academic success and well-being are available here: **https://acac.humboldt.edu/students/campus-student-support-services (https://acac.humboldt.edu/students/campus-student-support-services)**

Names and Genders

The roster of students enrolled in this course are provided by the Office of the Registrar and include your name as it appears on your official Cal Poly Humboldt student file. This name, however, may not be the name by which you want to be addressed. Please advise the instructor of your preferred name or gender pronoun, as early in the semester as possible so that changes made be made to the class records, and I can call you by the right name!

You'll find more background information and answers to frequently asked questions on the **Preferred Names & Pronouns (https://its.humboldt.edu/accounts-passwords/preferred-names)** page on the ITS

website.

Military Personnel

Veterans and currently serving military personnel and spouses with unique circumstances (upcoming deployments, drill/duty requirements, disabilities, VA appointments, etc.) are welcome and encouraged to communicate these, in advance if possible, to the instructor in the case that special arrangements need to be made.

Inclement Weather

In case of weather-related delays at the University, online portions of this course will proceed as planned. Your instructor will inform you if there are any extenuating circumstances regarding content or activity due dates in the course due to weather delays. If you are affected by a weather-related emergency, please contact your instructor at the earliest possible time to make special arrangements.

Additional Policies

Additional policy information is listed on the **Syllabus Addendum (https://www.humboldt.edu/academic-programs/syllabus-addendum)** page.

Course Copyright

All course materials students receive, or to which students have online access, are protected by copyright laws. Students may use course materials and make copies for their use as needed, but unauthorized distribution and uploading of materials without the instructor's express permission is strictly prohibited.

Students who engage in the unauthorized distribution of copyrighted materials may be held in violation of the University's Code of Conduct, and liable under Federal and State laws.

Disclaimer

Websites and other links may occasionally become unavailable, as is the fluid nature of the internet. While you should advise the instructor (by posting on Slack), it's in your best interest to search for alternate resources, should a link be broken, so that your progress is not delayed.

This syllabus is subject to change at any time, and you will be responsible for abiding by any such changes. Significant changes to the syllabus will be communicated to you via Canvas announcements.

Course Summary:

Date	Details	Due
Tue Jan 27, 2026	**Week 1 Mapping Assignment (https://canvas.humboldt.edu/courses/83688/assignments/744777)**	due by 11:59pm
Tue Feb 3, 2026	**Week 2 Mapping Assignment - Aggregation (https://canvas.humboldt.edu/courses/83688/assignments/761396)**	due by 11:59pm
Tue Feb 10, 2026	**Week 3 Reading Between the Lines (https://canvas.humboldt.edu/courses/83688/assignments/744751)**	due by 11:59pm
Mon Mar 16, 2026	**Spring Break (https://canvas.humboldt.edu/calendar?event_id=115435&include_contexts=course_83688)**	12am to Mar 21 at 12am
	Week 1 Reading Between the Lines (https://canvas.humboldt.edu/courses/83688/assignments/743187)	
	Week 2 Reading Between the Lines (https://canvas.humboldt.edu/courses/83688/assignments/743478)	